

- 5 (a) (i) Spreading of light waves when they pass through a small opening B1
- (ii) The laser light are coherent as their phase difference remains constant and does not vary with time. B1
- (b) (i) $d \sin \theta = n \lambda$
 $d \sin 68 = 3 \lambda$
 $d / \lambda = 3 / \sin 34$
- $\sin \theta_2 = \frac{2 \lambda}{d} = 2 \left(\frac{\sin 34}{3} \right)$ M1
- $2 \theta_2 = 44^\circ (43.78)$ A1
- (ii) $n = d / \lambda = 3 / \sin 34$ M1
 $= 3.23$
- Maximum $n = 5$ A1
- (iii) Since the wavelength of blue light is shorter, the angle will become smaller B1